IBM Z OMEGAMON AI for JVM Overview



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Overview

IBM® Z OMEGAMON® AI for JVM is a member of the OMEGAMON® family of mainframe monitoring products. You use OMEGAMON® AI for JVM to monitor the health of your vital Java™ virtual machines (JVMs) that run on z/OS® operating systems.

OMEGAMON AI for JVM assists you in anticipating performance problems and warns you when critical events take place in your site's JVMs by offering a central point of management for your Java application environments. OMEGAMON AI for JVM is an IBM Tivoli Monitoring (ITM) framework product which offers multiple user interfaces for different types of users.

Through OMEGAMON® AI for JVM, you can see operating and environmental characteristics about your JVMs, and key performance metrics, such as garbage collections rates, that help you manage the performance and availability of your JVMs. By using OMEGAMON® AI for JVM, you can quickly detect problems and take action in real time to speed up problem resolution.

With OMEGAMON® AI for JVM, you can monitor the performance of JVMs in different environments on z/OS, such as in CICS®, IMS™, or DB2® subsystems, or in pure Java™ application servers like the WebSphere® Application Server or WebSphere® Liberty Profile. OMEGAMON® AI for JVM agents that are installed on LPARs in your enterprise collect data from Java™ agents that are deployed on monitored JVMs. The Java™ agents provide JVM-level performance information that affects application response time and throughput such as garbage collection rates, general-purpose processor usage, and specialized processor usage.

OMEGAMON® AI for JVM can simultaneously monitor large numbers of JVMs running on the z/OS® operating system. It uses instrumentation that is built into IBM® JVM on z/OS® to discover key performance indicators and passes the metrics directly to the IBM® Z OMEGAMON® AI for JVM collector, where they are available for display in the Tivoli® Enterprise Portal and in the OMEGAMON® enhanced 3270 user interface.

The monitoring data that OMEGAMON® AI for JVM provides enables you to perform the following tasks:

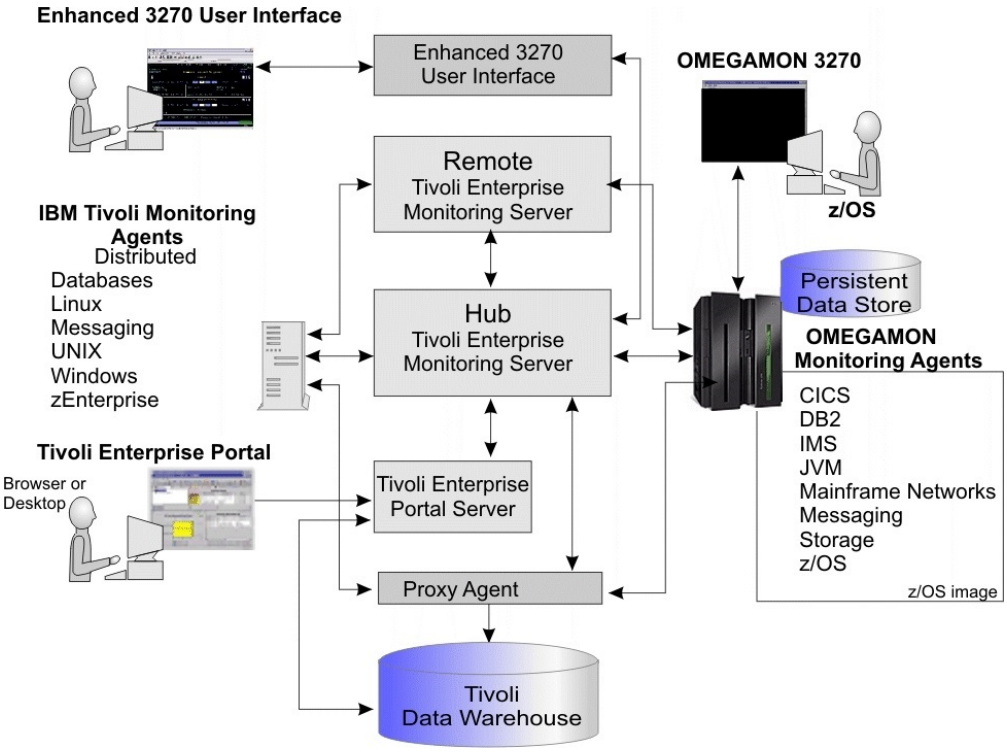
- Collect and analyze reliable, up-to-the-second data so that you can make fast and well-informed operating decisions
- Proactively manage the performance of all of your JVMs from a single, integrated interface
- Balance JVM workloads across various regions
- Track JVM performance against goals
- Exploit user-defined and predefined situations based on thresholds to raise different types of alerts
- Response time measurement for RESTful API providers and requesters hosted by IBM z/OS Connect Enterprise Edition (EE).

The Tivoli Management Services infrastructure

IBM® Z OMEGAMON® AI for JVM takes advantage of the Tivoli® Management Services infrastructure. Tivoli® Management Services components provide security, data transfer and storage, notification mechanisms, user interface presentation, and communication services for a number of products, including IBM® Tivoli® Monitoring and OMEGAMON® monitoring agents, in an agent-server-client architecture.

The following figure illustrates the relationship between Tivoli® Management Services components and monitoring agents. Java™ virtual machine (JVM) data that is collected by OMEGAMON® AI for JVM monitoring agents can be viewed in the OMEGAMON® enhanced 3270 user interface. You can also view the data through the graphical user interface called the Tivoli® Enterprise Portal.

Figure: Agent-server-client architecture



OMEGAMON® monitoring agents, including the OMEGAMON® AI for JVM monitoring agent, operate on monitored, or *managed*, systems. When you open a workspace in the Tivoli® Enterprise Portal or OMEGAMON® enhanced 3270 user interface (enhanced 3270UI), the workspace invokes a query, which is routed through the Tivoli® Management Services infrastructure to the appropriate monitoring agent. The monitoring agent collects the data, which is returned through the infrastructure and displayed in the Tivoli® Enterprise Portal or enhanced 3270UI client that requested it. For more information about Tivoli® Management Services components, refer to *OMEGAMON® shared documentation Version 6.3.0 Fix Pack 2 and above*.

IBM® Z OMEGAMON® AI for JVM components

Three components comprise IBM® Z OMEGAMON® AI for JVM: the Java™ agent, the collector, and the monitoring agent.

You must configure OMEGAMON® AI for JVM on each LPAR that has Java™ virtual machines (JVMs) that you want to monitor. The runtime environment includes the Java™ agent, which you deploy in each JVM that you want to monitor on the LPAR, a collector, and a monitoring agent.

Java™ agent

The IBM® Z OMEGAMON® AI for JVM Java agent runs in your monitored JVMs. The Java™ agent uses the IBM Z OMEGAMON AI JVM Tooling Interface (JVMTI) agent to collect performance metrics about the JVM in which it is running. The Java™ agent then sends the data to the IBM® Z OMEGAMON® AI for JVM collector. Each JVM that you want to monitor must be configured to run the Java™ agent, and you must also identify the collector to which data is sent.

Collector

The IBM® Z OMEGAMON® AI for JVM collector is a z/OS started task. It stores monitored JVM data that is passed to it from Java™ agents, and responds to requests for the JVM data by the OMEGAMON® AI for JVM monitoring agent.
The collector is identified by a unique ID.

Monitoring agent

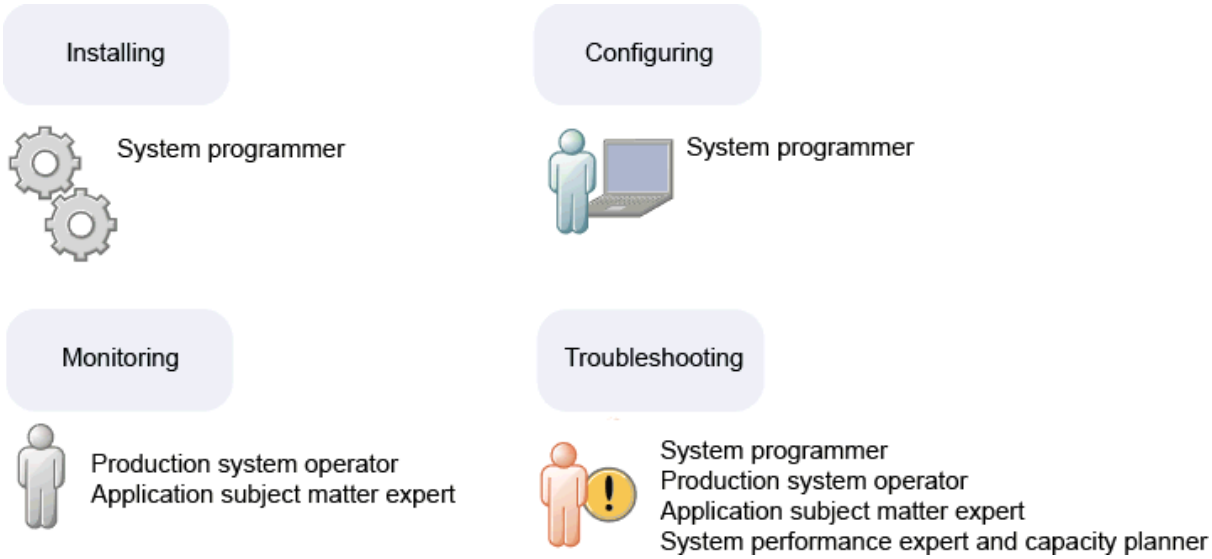
The OMEGAMON® AI for JVM monitoring agent is similar to the monitoring agents for other OMEGAMON® products in the OMEGAMON® family. It is a z/OS started task that responds to queries for JVM data from user interfaces. The OMEGAMON® AI for JVM monitoring agent communicates with the collector to obtain response data and returns the rows of data.
You must configure the monitoring agent to identify the collector with which it communicates.

The Java™ agent and collector exchange data by using cross-memory services. Data is also transferred between the collector and monitoring agent by using cross-memory services. The monitoring agent, similar to other

OMEGAMON® agents, uses either SNA or TCP/IP communication protocols to communicate with a Tivoli® Enterprise Monitoring Server.

User roadmap of OMEGAMON® AI for JVM

The following diagram depicts the OMEGAMON® AI for JVM user roadmap. Based on your role, you can click the relevant component in the diagram for detailed instructions.



User roles and responsibilities

System programmers

Typically, system programmers are responsible for the following tasks:

- Preparing and planning the environment deployment
- Installing and configuring OMEGAMON® AI for JVM in a runtime environment
- Configuring JVMs in applications and subsystems

Production system operators

Production system operators use OMEGAMON® AI for JVM to ensure consistent and reliable operations for all applications and subsystems on z/OS® that rely on Java™ technology. They are responsible for the following tasks:

- Taking diagnostic dumps for later analysis by application subject matter experts
- Reacting to real-time alerts when thresholds are breached, which indicates an adverse effect to an application
- Recycling the application in an orderly way while minimizing service interruption

Application subject matter experts

Application subject matter experts in Java™ and Java™ applications use OMEGAMON® AI for JVM during the development or integration testing to monitor the performance of the applications. By comparing the current and the previous versions of an application, application subject matter experts monitor differences in resources such as memory utilization, garbage collection, or locking conflicts.

System performance experts and capacity planners

System performance experts and capacity planners use OMEGAMON® AI for JVM to optimize system performance for all subsystems and applications. By using the performance metrics that OMEGAMON® AI for JVM provides for Java™ applications and JVMs that are embedded in subsystems, system performance experts and capacity planners are responsible for the following tasks:

- Creating situations and thresholds that are related to the applications under their control
- Ensuring that adequate resources are assigned to critical applications
- Identifying the systems that are wasting resources

Accessibility

Accessibility features help users with physical disabilities, such as restricted mobility or limited vision, to use software products successfully. OMEGAMON monitoring products support several user interfaces. Product functionality and accessibility features vary according to the interface.

The major accessibility features in this product enable users in the following ways:

- Use assistive technologies, such as screen-reader software and digital speech synthesizer, to hear what is displayed on the screen. Consult the product documentation of the assistive technology for details on using those technologies with this product.
- Operate specific or equivalent features using only the keyboard.
- Magnify what is displayed on the screen.

In addition, the product documentation was modified to include the following features to aid accessibility:

- All documentation is available in both HTML and convertible PDF formats to give the maximum opportunity for users to apply screen-reader software.
- All images in the documentation are provided with alternative text so that users with vision impairments can understand the contents of the images.

Some content presented in IBM Documentation might not yet be in a format that a screen reader can process. If you need help, contact ibmkc@us.ibm.com.

Interface information

The Tivoli Enterprise Portal interface offers the greatest range of functionality, but is not entirely accessible. The OMEGAMON enhanced 3270 user interface offers more limited functionality, but is entirely accessible. (The enhanced 3270 user interface supports all the accessibility features supported by your emulator. If you are using IBM Personal Communications, you can find information on its accessibility features at http://www.ibm.com/support/knowledgecenter/SSEQ5Y_6.0.0/com.ibm.pcomm.doc/books/html/quick_beginnings10.htm. If you are using a third-party emulator, see the documentation for that product for accessibility information.)
The OMEGAMON 3270 interface uses an ISPF style interface. Standard and custom PF Key settings, menu options, and command-line interface options allow for short cuts to commonly viewed screens. While basic customization options allow for highlights and other eye-catcher techniques to be added to the interface, the customization options are limited.

IBM and accessibility

See the IBM Human Ability and Accessibility Center for more information about the commitment that IBM has to accessibility.

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